How to Design ER Diagrams for E-commerce Website

In the domain of large-scale E-commerce websites, **data management** is important for smooth operational functioning. **Entity-Relation (ER) diagrams** constitute a fundamental tool for organizing and visualizing the relationships between entities within the system.

Designing the **e-commerce website** ER diagrams requires consideration of many parameters such as **scalability, flexibility**, and **performance optimization**. In this article, we'll explore the process of designing an **ER diagram** for **Large-Scale E-commerce websites**, focusing on key **entities**, **attributes**, **relationships**, and best practices for **database design**.

ER Diagram for E-Commerce Website

E-Commerce Website allows easy management of **products**, **orders**, **users**, **addresses**, **payments**, **tracking** information as well as **cart management**. **Products** are categorized in a user-friendly manner, as well as users can register and create multiple addresses in one account. **Orders** are placed by users.

Payment methods are diversified which is aimed at enhancing **security** and **flexibility** in **transactions**. **Tracking details** allow users to access their Order status from processing through to delivery. The **shopping cart** management system enables users to add, and check out products thereby helping with a smooth shopping process.

E-Commerce Website Features

Here are the features of Large-Scale E-commerce Platforms Features presented for the entities:

- **Product Management:** The e-commerce platform provides convenient product administration that gives users an easy way to insert, delete, edit, and view products. Specific information - product name, price, description, and category - is displayed, allowing the customer to make the right decisions.
- **Order Management:** Users can effortlessly make orders for products through the convenient design of the e-commerce platform. Every order

is assigned a distinct order number and the information attached includes the date of order, total amount, and other necessary details.

- **User Management:** This platform offers users account management features ensuring easy account creation with name and email.
- **Address Management:** Users can conveniently edit their shipping and billing addresses via the platform's address management module. The specification of a town, state, city, and zip code allows them to keep address details valid and inaccurate.
- **Payment Management:** The e-commerce platform provides a safe and convenient payment facility that allows users to finish transacting without difficulties. The user can select from different payment options which include UPI, credit cards, or others that are equally acceptable.
- **Tracking Order:** Users can also be aware of the status of their orders through the feature that is used to track orders. Our website ensures that each order is accompanied by multiple tracking details that customers can follow right from order processing to final delivery.
- **Cart Management:** The platform also has refined cart management feature. It is easy for a user to add some products to their shopping cart from the carts provided each of which can contain multiple products

Entities and Attributes for the E-commerce Website

Entities and Attributes are defined below:

1. Product:

- **P-ID(Primary Key):** Unique identifier for each product.
- **Name:** Name of the product.
- **Price:** Price of the product.
- **Description:** Description of the product.

2. Pro-category:

- **Category - ID(Primary Key):** Unique identifier for each category.
- **Name:** Name of the category.

3. Order:

- **Order - No(Primary Key):** Unique identifier for each order.
- **Order - Amount:**
- **Order - Date:**

User:

- **User - ID(Primary Key):** Unique identifier for each user or customer.
- **Name:** Name of the user.
- **Email:** Email of the user.

5. Address:

- **Address - ID(Primary Key):** Unique identifier for each address
- **Country:** Country of the user.
- **State:** State of the user.
- **City:** City of the user.
- **Pin - code:** Pin code of the user.

6. Payment:

- **Payment - ID(Primary Key):** Unique identifier for each payment.
- **Method:** Payment method like UPI or Credit Card etc.
- **Amount:** Total amount paid by user.

Tracking Detail:

- **Tracking - ID(Primary Key):** Unique identifier for each tracking detail.
- **Status:** Tracking status like on the way or delivered etc.
- **Order - No(Foreign Key):** Reference to the order which need to be tracked.

8. Cart:

- **Cart - ID(Primary Key):** Unique identifier for each cart.
- **User - ID(Foreign Key):** Reference to the user.

Relationships Between These Entities

1. Product - Prod-Category Relationship:

- One product can belong to only one category.
- One category can have multiple products.
- So this is a **Many-to-one** relationship, showing that many products can belong to a single category.

2. Order-User Relationship:

- One user can place multiple orders.
- Each order is placed by exactly one user.
- This is a **one-to-many** relationship, showing that a user can place multiple orders, but each order is placed by exactly one user.

3. User-Address Relationship:

- One user can have multiple addresses.
- Each address belongs to exactly one user.
- This is a **one-to-many** relationship, indicating that a user can have multiple addresses associated with their account.

4. Tracking Detail - Order Relationship:

- One order can have multiple tracking details.
- Each tracking detail corresponds to exactly one order.
- This is **Many-to-one** relationships showing that an order can have one or multiple associated tracking details.

5. Product - Cart Relationship:

- One product can be added to multiple carts.
- Each cart can contain multiple products.
- This is **many-to-many** relationship.

6. User-Payment Relationship:

User-Payment Relationship:

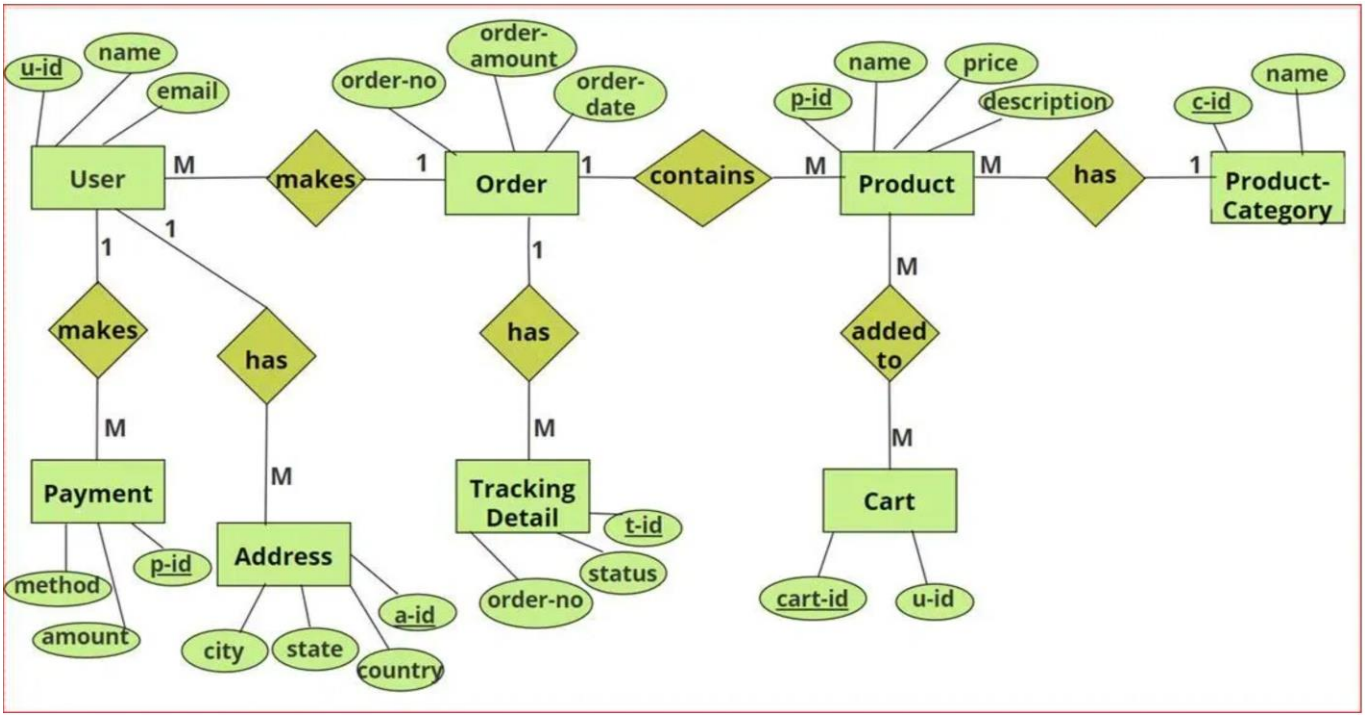
- One user can make multiple payments.
- Each payment is made by exactly one user.
- This is **one-to-many** relationship because each user can make multiple payments, and each payment is made by a single user.

7. Order-Product Relationship:

- One order can contains multiple products.
- Many products are get ordered in each order.
- So this is **one-to-many** relationship we can order multiple products on each order.

Representation of ER Diagram

Below is the ER diagram of large scale E-commerce platforms which meets all our requirements:


u-id
name
email
User
M
makes
1
order-no
order-amount
order-date
Order
1
contains
M
p-id
name
price
description
Product
M
has
1
c-id
name
Product-Category
1
1
makes
M
Payment
method
amount
p-id
has
M
Address
city
state
country
a-id
1
has
M
Tracking Detail
t-id
status
order-no
M
added to
M
Cart
cart-id
u-id